Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0234986945169713	1.0037549870922318 – 1.028179190751445

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.33792112617530035	0.3375684915229404 – 0.3384791160852235

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.47913481363078275	0.47624069058659213 – 0.48246535292720216

Thin QR

512 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.8726855191223079	0.8695658207967547 – 0.8755869170594145

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.6789324478702599	0.6782241501332267 – 0.6818595664936169

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9997738334754727	0.9953238460306846 – 1.0194679976178997

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9970730657484128	0.9917739551423491 – 1.0087548004946951

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9995848863933859	0.9948733492097857 – 1.0002446010470083

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.995085995085995 – 1.0003087372645878

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	**0.9988561419813765**	95% INTERVAL	0.9961922526300783 – 1.0001961530231342

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.002169477214626	1.0002808134228816 – 1.0036371291239314

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0001756182785322	0.9991833617518403 – 1.0009622780930691

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction

CURRENT / BASELINE	**0.9983208921448529**	95% INTERVAL	0.9957323002012497 – 0.9991901500488781

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	**0.9954545454545455**	95% INTERVAL	0.9889674681753889 – 1.0002854695974879

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.998486398278278	0.9957582184517497 – 1.003750007416945

Independent matrix operands

128 × 128 · 128 × 128 · f32

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0026257439607889	1.000093394662495 – 1.0043065616552176

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0041753653444678	1.0020876826722338 – 1.0062761506276152

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9995819397993311	0.9966555183946488 – 1

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0067263843058276	1.000258549180004 – 1.0133176513163846

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9993030841457505	0.9967016127769105 – 1.0029723941355255

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9962459765630284	0.994448678774562 – 0.999483314139962

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0091958049613556	0.9895708760248662 – 1.0212185378419087

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0036071534217699	0.997541024812381 – 1.006617502894442

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0003274059219547	0.9985314514155177 – 1.0022661848471857